

When Boolean logic is not too elegant

Consider the following argument:

Everyone in this room is feeling miserable today.

Therefore, Agi is feeling miserable today.

We can formalise this argument by choosing propositional variables

p : “James is feeling miserable today.”

q : “Sandeep is feeling miserable today.”

r : “Agi is feeling miserable today.”

z : “Zoe is feeling miserable today.”

...

Then we can translate, say, the assumption above as

$$p \wedge q \wedge r \wedge z \wedge \dots$$

and then the conclusion r logically follows.

This is rather cumbersome, but adequate. There are only finitely many people in the room, so $p \wedge q \wedge r \wedge z \wedge \dots$ (if we write it out fully) is a formula.

When Boolean logic is not enough

Consider the following argument:

- (1) *Every even number, except two, is not a prime.*
- (2) *Six is an even number and six is not two.*
- (3) *Hence, six is not a prime.*

We cannot use the 'trick' from the previous slide, as there are *infinitely* many numbers, but Boolean logic does allow *only finite* conjunctions.

- We can use **predicate logic** instead:

$$(1): \forall x (\text{EVEN}(x) \wedge \neg(x = 2) \rightarrow \neg\text{PRIME}(x))$$

$$(2): \text{EVEN}(6) \wedge \neg(6 = 2)$$

$$(3): \neg\text{PRIME}(6)$$

De Morgan's famous example

A horse is an animal.

Therefore, the head of a horse is a head of an animal.

A translation in predicate logic:

$$\forall x (\text{HORSE}(x) \rightarrow \text{ANIMAL}(x))$$

$$\forall y (\exists x (\text{HORSE}(x) \wedge \text{head}(x) = y) \rightarrow \exists x (\text{ANIMAL}(x) \wedge \text{head}(x) = y))$$

Predicate logic uses all the machinery of Boolean logic, plus:

- predicate symbols (such as EVEN, PRIME, HORSE),
- constants and function symbols (such as 6, head), and
- quantifiers over individual variables ($\forall y$ and $\exists x$).

Why to consider more kinds of logical models

- In *Boolean logic* all we can say about sentences that they are either true or false, in any given interpretation. Boolean logic does not allow for any further analysis of arguments, which may turn out to be inadequate from many points of view.
- *Predicate logic* provides us with a more detailed analysis of sentences and logical arguments. The price we pay for its increased expressiveness is its complex nature and undecidability:

There are NO effective, programmable methods for deciding validity, satisfiability, or logical consequence in predicate logic.

- Anything else?

Modal logics

- **Modal logics** are logical formalisms that are designed for handling different modes of truth (**'modalities'**).
- There are many different kinds of modal logics: epistemic, temporal, dynamic, deontic, Each of them is tailored to describe and reason about different notions such as necessity, possibility, knowledge, belief, obligation, change in time, behaviour of programs, etc.
- Though all these features can be handled by predicate logic as well, modal logic gives a useful alternative to predicate logic.
 - For several application areas, it is more natural and intuitive to work with modal logic than with predicate logic.
 - In many cases, the logical procedures for modal logics are **effective** (that is, decidable, programmable), unlike for predicate logic.

Modes of truth

In natural language we often distinguish between various 'modes' of truth.

For example:

(1) *'Barack Obama is the president of the USA.'*

This is currently true, but it will not be **true some time in the future**.

(2) *'There are ten planets in our solar system.'*

While this is true(?), and maybe true forever in the future, is not **necessarily true** in the sense that it could have been a different number.

Modes of truth II

(3) ' $2 + 3 = 5$.'

This is true now and always in the future, and necessarily true in the sense that it follows from the laws of mathematics. But it does not enjoy all modes of truth: It may not be **known to be true** by e.g. small children; it may not be **believed to be true** by some people.

(4) '*It ought to be the case that we wear a seat-belt in a car.*'

This is only true depending on what **moral** codes we adopt.

(5) '*The value of the integer variable x is 19.*'

This is **true after some execution of a program** P (while might not be true for other executions of the same program).

Basic modal logic: alphabet

As before, we start describing a logic with defining its formal language (*syntax*).

The alphabet of basic modal logic consists of the following symbols:

- the symbols in the alphabet of Boolean logic (see slide 21)
- plus two new unary logical connectives:
 - (pronounced as 'box') and
 - ◇ (pronounced as 'diamond')

(Usually, □ and ◇ are called modal connectives or modalities.)

Syntax of basic modal logic: formulas

Well-formed **formulas** of basic modal logic (aka **modal formulas**) are built up inductively as follows:

- *Base case:* Any propositional variable is a modal formula.
- *Inductive step:* If A and B are modal formulas then so are

$$(\neg A) \quad (A \wedge B) \quad (A \vee B) \quad (A \rightarrow B) \quad (A \leftrightarrow B) \quad (\Box A) \quad (\Diamond A)$$

- *Closure:* Nothing else is a modal formula.

Examples:

- Any formula of Boolean logic is also a modal formula.
- $((\Box p) \wedge (\Box(\neg q))) \rightarrow r$

Notational conventions

- We omit the outermost parentheses of formulas.
- $\Box$, $\Diamond$ and $\neg$ bind more tightly than $\vee$, $\wedge$ and $\rightarrow$, so we omit the parentheses in occurrences of $(\Box A)$, $(\Diamond A)$ and $(\neg A)$.

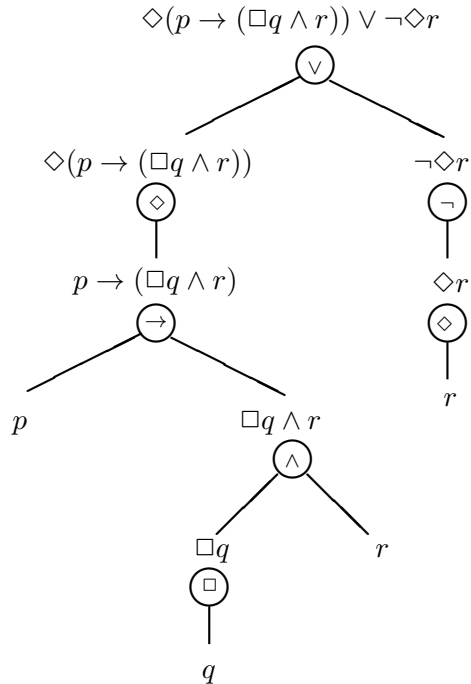
Examples:

we write $(\Diamond p \wedge \Box \neg q) \rightarrow r$ for $((\Diamond p) \wedge (\Box(\neg q))) \rightarrow r$

we write $\Box(p \wedge q) \vee (\Diamond r \wedge \neg \Box p)$ for $(\Box(p \wedge q)) \vee ((\Diamond r) \wedge (\neg(\Box p)))$

But always use brackets if in any doubt!

Parse tree of a formula



Any well-formed modal formula can be decomposed this way.

The formulas at the nodes of the parse tree are the **subformulas** of the initial formula.

Various readings of $\Box$ and $\Diamond$

$\Box A$:	$\Diamond A$:
I know A	A is compatible with my knowledge
A is necessary	A is possible
A will always be true	A will be true some time
A is obligatory	A is permitted
I believe A	A is compatible with my beliefs
A holds after any execution of my program P	A holds after some execution of my program P
...	...

Epistemic reading

$\Box A$ reads as '*I know A*' or '*A is known*' or '*agent Q knows A*'.

Modal formula	Reading
$\Box\Box p$	p is known to be known.
$\Box p \rightarrow p$	If p is known then p is true.
$\Box p \rightarrow \Diamond p$	If Q knows p then p is consistent with Q 's knowledge.
$\Box p \vee \Box \neg p$	Either Q knows p or Q knows not p .
$\Box(p \rightarrow q) \rightarrow (\Box p \rightarrow \Box q)$	? <i>logical omniscience</i>
$\Box p \rightarrow \Box\Box p$	? <i>positive introspection</i>
$\neg\Box p \rightarrow \Box\neg\Box p$	? <i>negative introspection</i>

Deontic reading

deontic = dealing with norms.

$\Box A$ reads as '*A is obligatory*' or '*A is ought to be true*',

$\Diamond A$ reads as '*A is permitted*'.

Modal formula	Reading
$(\Box p \wedge \Box q) \rightarrow \Box(p \wedge q)$	If both p and q are obligatory then so is ' p and q together'.
$\Box p \rightarrow \Diamond p$	If p is obligatory then p is permitted.
$p \rightarrow \Box p$	If p is true then p is obligatory.

Temporal reading

$\Box A$ reads as '*A will always be true in the future*' ,

$\Diamond A$ reads as '*A will be true some time in the future*' .

Modal formula	Reading
$\Box p \rightarrow p$	If p will always be true then p is true now.
$\Box(p \rightarrow \Box p)$	Once p , always p .
$\Diamond \Box p$	Eventually permanently p .
$\Box(p \rightarrow \Diamond q)$	Whenever p , there is always a q afterwards.
$\Box \Diamond 1$	Time has no end.
$\Box 0$	Now it is the end of time.

Executions of a program

By an 'execution' of a program here we mean a *terminating* execution.

$\Box A$ reads as '*A holds after every execution of a program P*'.

$\Diamond A$ reads as '*A holds after some execution of a program P*'.

Modal formula	Reading
$(\Diamond p \wedge \Box q) \rightarrow \Diamond(p \wedge q)$	If there is an execution ending with p true, and every execution ends up with q true, then there is an execution with both p and q true.
$(\Diamond p \wedge \Diamond q) \rightarrow \Diamond(p \wedge q)$	If there is an execution with p true, and (possibly another) one with q true, then there is an execution with both p and q true.
$\Box(p \wedge \neg p)$	The program never terminates.

Multi-modal logics

Sometimes the setting we want to model needs *more than one* modality.

- **Multi-agent systems:** Different agents $Q_1, Q_2, \dots$ have different knowledge of the world. We need different modalities $\Box_1, \Box_2, \dots$ and $\Diamond_1, \Diamond_2, \dots$ to model this situation. Then we can have modal formulas like $\Box_1 \Box_2 p \rightarrow \Box_2 \Diamond_1 p$.
- **Reasoning about time:** We may want to speak about things in the past, not only in the future. Then we need a box modality $\Box_F$ for '*always in the future*' and another one $\Box_P$ for '*always in the past*', plus their diamond-variants $\Diamond_F$ and $\Diamond_P$.
- **Reasoning about programs:** In order to speak about the behaviour of different programs at the same time, we need different modalities $\Box_{P_1}, \Box_{P_2}, \dots$, and $\Diamond_{P_1}, \Diamond_{P_2}, \dots$, for each program $P_1, P_2, \dots$. This makes the logic suitable to speak about complex processes that are put together from simple parts.

Necessity

What if $\Box A$ reads as '*A is necessarily true*' ?

There are different kinds of necessities. For example, let p denote the statement

'The velocity of every material body is smaller than that of light.'

- Then the modal formula $\Box p$ expresses a true statement if $\Box$ is read as **physical necessity**: It follows from the laws of physics that p is true.
- However, if $\Box$ reads as **logical necessity**, then it is debatable whether $\Box p$ holds: It is logically possible that p is not true.
- Moreover, there are modal formulas, such as $\Box q \rightarrow \Box q$, that seem to be true *whatever* reasonable meaning we give to $\Box$.

~> **As before, we need a precise formulation of meaning.**

semantics: 'model circumstances'

‘Possible world’ semantics: the idea

Leibniz’ idea: necessity = truth in **all** possible worlds

possibility = truth in **at least one** possible world

So, we can imagine a system of possible *worlds* in which each world has some (possibly none) *alternatives*.

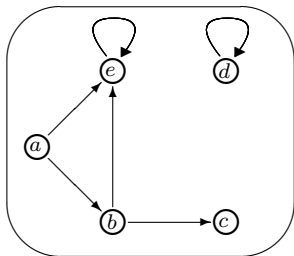
- Every world ‘lives under the laws’ of Boolean logic:
Each world has its own interpretation of the propositional variables, each of them are either true (**1**) or false (**0**) in it. The truth-values of compound formulas that are composed with the help of the Boolean logical connectives are determined by the usual truth-tables.
- Different worlds might have different interpretations of the same variables.
- A formula of the form $\Box A$ is regarded to be true in a world w , if A is true in **all** the worlds alternative to w .
- A formula of the form $\Diamond A$ is true in a world w , if A is true in **at least one** alternative of w .

Kripke frames

A Kripke frame is a pair $\mathfrak{F} = (W, R)$ where

- W is a non-empty set of worlds (or states, or points)
- R is a binary relation on W , the alternativeness (or accessibility) relation.

For example:



$$\mathfrak{F} = (W, R)$$

$$W = \{a, b, c, d, e\}$$

$$R = \{(a, b), (a, e), (b, c), (b, e), (d, d), (e, e)\}$$

If $(a, b) \in R$, then we say: ' a **sees** b ', or

' b **is accessible from** a ', or

' b **is an alternative for** a '.

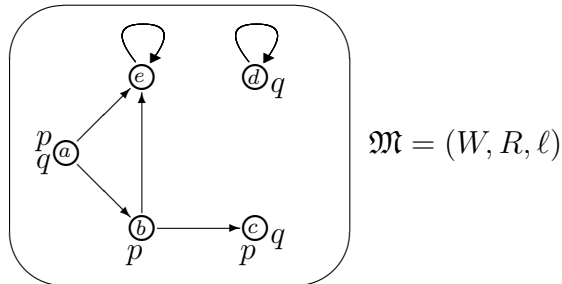
we also write: aRb

‘Model circumstances’ for modal logic: Kripke models

A Kripke model is a triple $\mathfrak{M} = (W, R, \ell)$ where

- (W, R) is a Kripke frame, and
- ℓ is a function labelling each world with the propositional variables that are supposed to be true in it. Such a labelling function ℓ is called a valuation.

For example:



- We say that model $\mathfrak{M} = (W, R, \ell)$ is based on the frame (W, R) .
- There can be many different models based on the same frame.

The meaning of modal formulas in Kripke models

Now we ‘connect’ the formal language and the ‘model circumstances’ of basic modal logic: We describe what modal formulas ‘mean’ in Kripke models.

For every modal formula A , Kripke model $\mathfrak{M}$ and world w in $\mathfrak{M}$, we will define the notions:

“ A is true in $\mathfrak{M}$ at world w ”

notation:

$$\boxed{\mathfrak{M} \models_w A}$$

“ A is false in $\mathfrak{M}$ at world w ”

$$\boxed{\mathfrak{M} \not\models_w A}$$

(There are other notations for this in the literature:

$(\mathfrak{M}, w) \models A$, or $\mathfrak{M}, w \Vdash A$, or $\|A\|_w^{\mathfrak{M}} = 1$, ...)

Evaluating modal formulas in Kripke models

Slogan: Every world w ‘lives under the laws’ of Boolean logic. So the ‘behaviour’ of Boolean logic connectives is determined by their usual truth-tables.

- If p is a propositional variable, then $\boxed{\mathfrak{M} \models_w p}$ if p is among the labels of w .
If p is not among the labels of w , then $\boxed{\mathfrak{M} \not\models_w p}$.
- $\boxed{\mathfrak{M} \models_w \neg A}$ if $\mathfrak{M} \not\models_w A$. Otherwise, $\boxed{\mathfrak{M} \not\models_w \neg A}$.
- $\boxed{\mathfrak{M} \models_w A \wedge B}$ if $\mathfrak{M} \models_w A$ and $\mathfrak{M} \models_w B$. Otherwise, $\boxed{\mathfrak{M} \not\models_w A \wedge B}$.
- The cases of $A \vee B$, $A \rightarrow B$, and $A \leftrightarrow B$ are similar, see their truth-tables:

A	B	$A \wedge B$	$A \vee B$	$A \rightarrow B$	$A \leftrightarrow B$
1	1	1	1	1	1
1	0	0	1	0	0
0	1	0	1	1	0
0	0	0	0	1	1

Evaluating modal formulas in Kripke models II

- A formula of the form $\Diamond A$ is regarded to be **true** at a world w , if A is true in **at least one** alternative of w :

$\mathfrak{M} \models_w \Diamond A$ if there is a world u in $\mathfrak{M}$ such that wRu and $\mathfrak{M} \models_u A$.

- Otherwise, $\mathfrak{M} \not\models_w \Diamond A$.

So, a formula of the form $\Diamond A$ is regarded to be **false** at a world w , if A is false at all the worlds alternative to w .

$\Diamond$ is a bit like $\exists$ of predicate logic

Special case: If world w is a 'dead-end' (a world with no alternatives), then $\Diamond A$ is **always false at** w (no matter what A is).

Evaluating modal formulas in Kripke models III

- A formula of the form $\Box A$ is regarded to be **true** at a world w , if A is true at **all** the worlds alternative to w :

$$\boxed{\mathfrak{M} \models_w \Box A} \quad \text{if} \quad \mathfrak{M} \models_u A \text{ for all worlds } u \text{ in } \mathfrak{M} \text{ with } wRu.$$

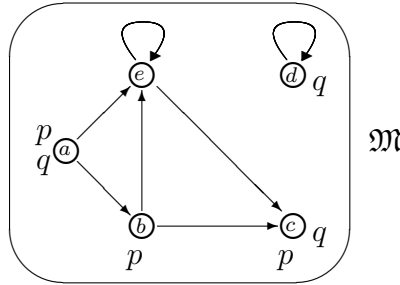
- Otherwise, $\boxed{\mathfrak{M} \not\models_w \Box A}$.

So, a formula of the form $\Box A$ is regarded to be **false** at a world w , if there is **at least one** world alternative to w where A is false.

$\Box$ is a bit like $\forall$ of predicate logic

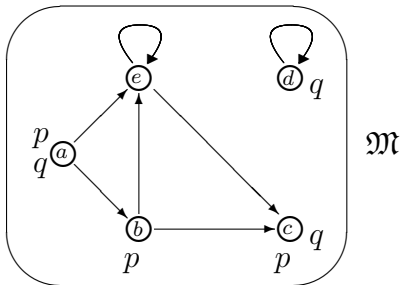
Special case: If world w is a 'dead-end' (a world with no alternatives), then $\Box A$ is **always true at** w (no matter what A is).

Exercise 5.1: model checking in Kripke models



- (1) $\mathfrak{M} \models_a p$: It is because p is among the labels of a .
- (2) $\mathfrak{M} \models_a \Box \neg q$: a has two alternatives, b and e . q is not among the labels of either of them, so $\mathfrak{M} \not\models_b q$ and $\mathfrak{M} \not\models_e q$, hence $\mathfrak{M} \models_b \neg q$ and $\mathfrak{M} \models_e \neg q$.
- (3) $\mathfrak{M} \not\models_e \Box q$: $e R e$ and $\mathfrak{M} \not\models_e q$.
- (4) $\mathfrak{M} \models_a \Diamond \Diamond (p \wedge q)$: $a R b$ and $b R c$, and $\mathfrak{M} \models_c p \wedge q$.
- (5) $\mathfrak{M} \not\models_a \Diamond \Diamond (p \wedge q) \wedge \Diamond q$: a has two alternatives, b and e , and q is false in both. So $\mathfrak{M} \not\models_a \Diamond q$.
- (6) $\mathfrak{M} \not\models_c \Diamond 1$: There is no world accessible from c , so all formulas of the form $\Diamond A$ are false in c .
- (7) $\mathfrak{M} \models_c \Box 0$: There is no world accessible from c , so all formulas of the form $\Box A$ are true in c (because we can't find a 'witness' to make them false).

Exercise 5.1: model checking in Kripke models (cont.)



- (8) $\mathfrak{M} \models_d \Diamond \Diamond q$: dRd and dRd , and $\mathfrak{M} \models_d q$.
- (9) $\mathfrak{M} \models_b \Diamond \Box \neg q$: bRc and c sees no other worlds, so all $\Box$ -formulas are true in c . In particular, $\mathfrak{M} \models_c \Box \neg q$.
- (10) $\mathfrak{M} \models_b \Diamond \Box q$: bRc and c sees no other worlds, so all $\Box$ -formulas are true in c . In particular, $\mathfrak{M} \models_c \Box q$.
- (11) $\mathfrak{M} \models_d \Diamond p \vee \Box q$: The only alternative of d is d , and $\mathfrak{M} \models_d q$, so $\mathfrak{M} \models_d \Box q$.
- (12) $\mathfrak{M} \not\models_e \Box(\Box q \vee \Box p)$: eRe , $\mathfrak{M} \not\models_e p$ and $\mathfrak{M} \not\models_e q$, so $\mathfrak{M} \not\models_e \Box p \vee \Box q$.

Truth and satisfiability

- A formula A is **true in a model** $\mathfrak{M}$ if

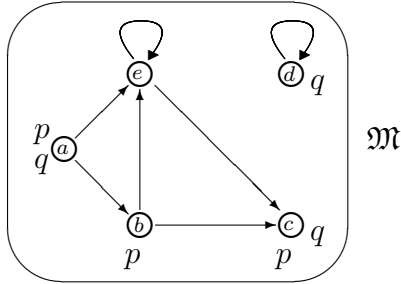
$$\mathfrak{M} \models_w A \text{ for all worlds } w \text{ in } \mathfrak{M}.$$

- A formula A is **satisfiable in a model** $\mathfrak{M}$ if

there is some world w in $\mathfrak{M}$ such that $\mathfrak{M} \models_w A$.

If A is true in $\mathfrak{M}$ then A is satisfiable in $\mathfrak{M}$.
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Exercise 5.2: Truth and satisfiability



- (1) $\Box \neg p \wedge \Box \Box \neg p$: Satisfiable in $\mathfrak{M}$, as $\mathfrak{M} \models_c \Box \neg p \wedge \Box \Box \neg p$.
Not true in $\mathfrak{M}$, as $\mathfrak{M} \not\models_a \Box \neg p \wedge \Box \Box \neg p$.
- (2) $\Diamond(\neg p \vee \Diamond q)$: Satisfiable in $\mathfrak{M}$, as $\mathfrak{M} \models_a \Diamond(\neg p \vee \Diamond q)$.
Not true in $\mathfrak{M}$, as $\mathfrak{M} \not\models_c \Diamond(\neg p \vee \Diamond q)$.
- (3) $\Box p \rightarrow p$: True in $\mathfrak{M}$, as p is true in a, b and c , and $\Box p$ is false in d and e .
- (4) $\Box(p \leftrightarrow p)$: True in $\mathfrak{M}$.
Can you find a model where this formula is *not* true?